

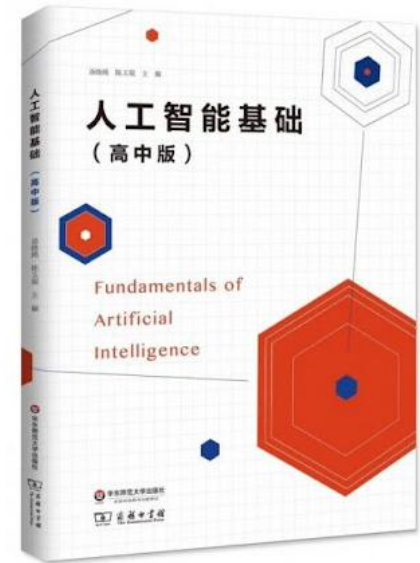
UGEB1407 Demystifying AI

Lecture 1: Introduction

The World of AI

Excerpt from an AI textbook for high-school students:

- In the morning, your smart-home system says “good morning” using human voice, and you speak to open the window/curtain, and ask about the weather forecast
- An intelligent robot prepares the breakfast according to your health data and dining history. When having breakfast, you watch personalized news information
- After breakfast time, a self-driving car moves to the front door of your house. The sensing system of the car can recognize you and allow you to get in the car
- The car picks up the fastest route according to real-time traffic information and takes you to your work place
- In the afternoon, you visit a hospital for medical imaging examination. The medical images are analysed by computers and you get the results and clinical recommendation in 10 minutes
- You go to buy new clothes. The shop provides intelligent fitting system which allows you to view your 3D image of wearing a selected cloth



Artificial Intelligence in movies



https://youtu.be/_19pRsZRiz4

Artificial Intelligence in movies



<https://www.youtube.com/watch?v=gyKqH0gMi4g>

Seriously ?





AI Technology Invades Hair Tools and Appliances

The hair tools and appliances market has finally been bitten by the Artificial Intelligence (AI) bug.



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Ⓢ A.I. 級報價

個人化報價，度身訂做，準確批足[^]。



Ⓢ A.I. 級解答

24/7 chatbot，即解答疑難。

申請





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首页

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https://www.youtube.com/watch?v=AuBs954_Rrw

Ping An Good Doctor Upgrades AI Doctor Service “Ping An AI Doctor”, Expanding Access to Ping An Ecosystem’s 90 Million MAUs

By PR Newswire



<https://macaubusiness.com/ping-an-good-doctor-upgrades-ai-doctor-service-ping-an-ai-doctor-expanding-access-to-ping-an-ecosystems-90-million-maus/>

INTRODUCTION

One of the obligations that comes with my work as a venture-capital (VC) investor is that I often give speeches about artificial intelligence (AI) to members of the global business and political elite. One of the joys of my work is that I sometimes get to talk about that very same topic with kindergarteners. Surprisingly, these two distinctly different audiences often ask me the same kinds of questions. During a recent visit to a Beijing kindergarten, a gaggle of five-year-olds grilled me about our AI future.

"Are we going to have robot teachers?"

"What if one robot car bumps into another robot car and then we get hurt?"

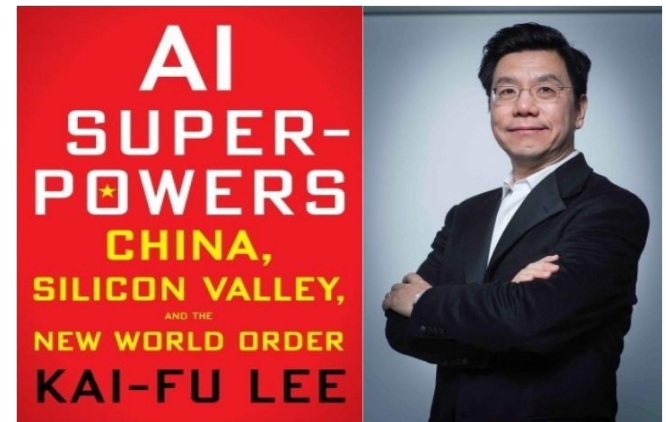
"Will people marry robots and have babies with them?"

"Are computers going to become so smart that they can boss us around?"

"If robots do everything, then what are we going to do?"

These kindergarteners' questions echoed queries posed by some of the world's most powerful people, and the interaction was revealing in several ways. First, it spoke to how AI has leapt to the forefront of our minds. Just a few years ago, artificial intelligence was a field that lived primarily in academic research labs and science-fiction films. The average person may have had some sense that AI was about building robots that could think like people, but there was almost no connection between that prospect and our everyday lives.

Today all of that has changed. Articles on the latest AI innovations blanket the pages of our newspapers. Business conferences on



ARTIFICIAL INTELLIGENCE

~~__MACHINE LEARNING__~~

~~__BIG DATA~~ IS LIKE **TEENAGE
SEX: EVERYONE** TALKS
ABOUT IT, **NOBODY** REALLY
KNOWS HOW TO DO IT,
EVERYONE THINKS
EVERYONE ELSE IS DOING IT,
SO **EVERYONE** CLAIMS THEY
ARE DOING IT...

<https://medium.com/point-nine-news/18-geeky-predictions-for-2016-af891f24ec12>

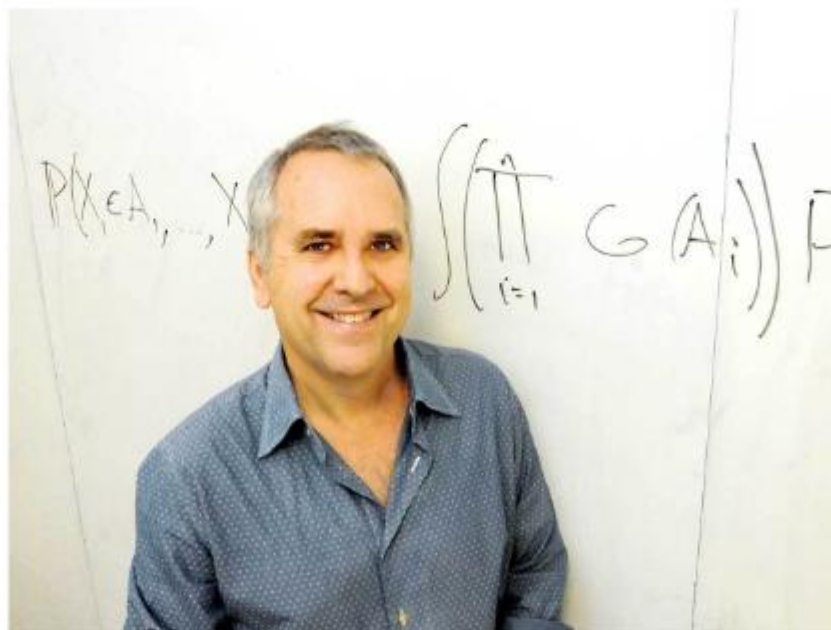
When there is something seemingly involving “human intelligence” that can be done by a computer (or any automatic machine) and people find it not easy to understand how it is done, they would say “This must be AI”.

[Article](#) | [The Institute](#) | [IEEE Member News](#)

Stop Calling Everything AI, Machine-Learning Pioneer Says

Michael I. Jordan explains why today's artificial-intelligence systems aren't actually intelligent

By **Kathy Pretz**



Interview

Microsoft's Kate Crawford: 'AI is neither artificial nor intelligent'

Zoë Corbyn



“AI is neither artificial nor intelligent. It is made from natural resources and it is people who are performing the tasks to make the systems appear autonomous.”

So, what is AI ?

How CEO learn AI



Executive Education

English

About Programme



AI & Big Data Experiential Workshop for Executives

Berkeley ExecEd | Berkeley Haas



Artificial Intelligence: Business Strategies and Applications

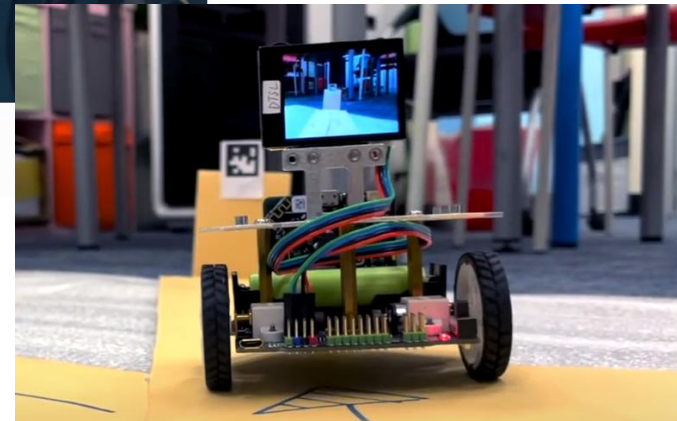
Lead AI Initiatives
with Confidence

PREVIEW PROGRAM FOR FREE

How kids learn AI



AI for the Future Summer Camp 2026



How serious technologists do AI



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Reinforcement-Learning-Algorithms-with-Python / Chapter05 / DQN_Atari.py / <> Jump to ▾

Go to file

⋮



casijoe5231 Code files updated

Latest commit 53bd33e on 22 Oct 2019 🕒 History

👤 1 contributor

337 lines (254 sloc) | 11.5 KB

Raw

Blame



```
1 import numpy as np
2 import tensorflow as tf
3 import gym
4 from datetime import datetime
5 from collections import deque
6 import time
7 import sys
8
9 from atari_wrappers import make_env
```

10

11

What serious AI researchers do

Cornell University

arXiv.org > stat > arXiv:1406.2661

Search... All fields Search

Help | Advanced Search

Statistics > Machine Learning

[Submitted on 10 Jun 2014]

Generative Adversarial Networks

Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $1/2$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

Subjects: Machine Learning (stat.ML); Machine Learning (cs.LG)

Cite as: arXiv:1406.2661 [stat.ML]
(or arXiv:1406.2661v1 [stat.ML] for this version)

Submission history

From: Ian Goodfellow [view email]

[v1] Tue, 10 Jun 2014 18:58:17 UTC (1,257 KB)

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References & Citations

- NASA ADS
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50 blog links (what is this?)

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Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**
 for k steps **do**
 • Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
 • Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
 • Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log(1 - D(G(z^{(i)}))) \right].$$

end **for**
• Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
• Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log(1 - D(G(z^{(i)}))).$$

end **for**
The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

Generative Adversarial Nets

Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio¹
Département d'informatique et de recherche opérationnelle
Université de Montréal
Montréal, QC H3C 3J7

Abstract

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $1/2$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

1 Introduction

The promise of deep learning is to discover rich, hierarchical models [2] that represent probability distributions over the kinds of data encountered in artificial intelligence applications, such as natural images, audio waveforms containing speech, and symbols in natural language corpora. So far, the most striking successes in deep learning have involved discriminative models, usually those that map a high-dimensional, rich sensory input to a class label [14, 22]. These striking successes have primarily been based on the backpropagation and dropout algorithms, using piecewise linear units [19, 9, 10] which have a particularly well-behaved gradient. Deep generative models have had less of an impact, due to the difficulty of approximating many intractable probabilistic computations that arise in maximum likelihood estimation and related strategies, and due to difficulty of leveraging the benefits of piecewise linear units in the generative context. We propose a new generative model estimation procedure that sidesteps these difficulties.¹

In the proposed *adversarial nets* framework, the generative model is pitted against an adversary: a discriminative model that learns to determine whether a sample is from the model distribution or the data distribution. The generative model can be thought of as analogous to a team of counterfeiters, trying to produce false currency and use it without detection, while the discriminative model is analogous to the police, trying to detect the counterfeit currency. Competition in this game drives both teams to improve their methods until the counterfeiters are indistinguishable from the genuine articles.

¹Alex Pouget-Abadie is visiting Université de Montréal from Ecole Polytechnique.

²Sherjil Ozair is visiting Université de Montréal from Indian Institute of Technology Delhi

³Yoshua Bengio is a CIFAR Senior Fellow

⁴All code and hyperparameters available at <http://www.github.com/sgoodfellow/adversarial>.

This framework can yield specific training algorithms for many kinds of model and optimization algorithm. In this article, we explore the special case when the generative model generates samples by passing random noise through a multilayer perceptron, and the discriminative model is also a multilayer perceptron. We refer to this special case as *adversarial nets*. In this case, we can train both models using only the highly successful backpropagation and dropout algorithms [17] and sample from the generative model using only forward propagation. No approximate inference or Markov chains are necessary.

2 Related work

An alternative to directed graphical models with latent variables are undirected graphical models with latent variables, such as restricted Boltzmann machines (RBMs) [27, 16], deep Boltzmann machines (DBMs) [26] and their numerous variants. The interactions within such models are represented as the product of unnormalized potential functions, normalized by a global summation/integration over all states of the random variables. This quantity (the *partition function*) and its gradient are intractable for all but the most trivial instances, although they can be estimated by Markov chain Monte Carlo (MCMC) methods. Mixing poses a significant problem for learning algorithms that rely on MCMC [3, 5].

Deep belief networks (DBNs) [16] are hybrid models containing a single undirected layer and several directed layers. While a fast approximate layer-wise training criterion exists, DBNs incur the computational difficulties associated with both undirected and directed models.

Alternative criteria that do not approximate or bound the log-likelihood have also been proposed, such as score matching [18] and noise-contrastive estimation (NCE) [13]. Both of these require the learned probability density to be analytically specified up to a normalization constant. Note that in many interesting generative models with several layers of latent variables (such as DBNs and DBMs), it is not even possible to derive a tractable unnormalized probability density. Some models such as denoising auto-encoders [30] and contractive auto-encoders have learning rules very similar to score matching applied to RBMs. In NCE, as in this work, a discriminative training criterion is employed to fit a generative model. However, rather than fitting a separate discriminative model, the generative model itself is used to discriminate generated data from samples a fixed noise distribution. Because NCE uses a fixed noise distribution, learning slows dramatically after the model has learned even an approximately correct distribution over a small subset of the observed variables.

Finally, some techniques do not involve defining a probability distribution explicitly, but rather train a generative machine to draw samples from the desired distribution. This approach has the advantage that such machines can be designed to be trained by back-propagation. Prominent recent work in this area includes the generative stochastic network (GSN) framework [5], which extends generalized denoising auto-encoders [42] both can be seen as defining a parameterized Markov chain, i.e., one learns the parameters of a machine that performs one step of a generative Markov chain. Compared to GSNs, the adversarial nets framework does not require a Markov chain for sampling. Because adversarial nets do not require feedback loops during generation, they are better able to leverage piecewise linear units [19, 9, 10], which improve the performance of backpropagation but have problems with unbounded activation when used in feedback loop. More recent examples of training a generative machine by back-propagating into it include recent work on auto-encoding variational Bayes [20] and stochastic backpropagation [24].

3 Adversarial nets

The adversarial modeling framework is most straightforward to apply when the models are both multilayer perceptrons. To learn the generator's distribution p_g over data x , we define a prior on input noise variables $p_z(z)$, then represent a mapping to data space as $G(z; \theta_g)$, where G is a differentiable function represented by a multilayer perceptron with parameters θ_g . We also define a second multilayer perceptron $D(x; \theta_d)$ that outputs a single scalar. $D(x)$ represents the probability that x came from the data rather than p_g . We train D to maximize the probability of assigning the correct label to both training examples and samples from G . We simultaneously train G to minimize $\log(1 - D(G(z)))$.

In this course, we will

1. **discuss** what is meant by AI, yesterday, today and in future;
2. **explain** what is inside today's AI technology and how a typical AI system works
3. **analyze** existing AI applications and understand the limitations
4. **explore** potential of AI in a wide range of areas
5. **consider** societal and ethical issues related to AI

Perspectives adopted in the course

People are interested in AI from different perspectives.

Questions about AI could go very practical, technical, or philosophical.

In this course, our discussion will be mainly from the perspectives of technology:

- What is it ?
- How does it work ?
- What can be done ?
- What cannot be done ?
- How will it evolve ?

Expected outcomes

After taking the course, students

- can develop a more educated perspective on AI and related issues
- would not be misled or fooled by hearsay and overtalk about AI
- can more easily understand upcoming new development of AI
- would take the courage to consider using AI in their areas of specialization

Artificial ... Intelligence ...

From Oxford English Dictionary:

Artificial -- made or constructed by human skill, esp. in imitation of, or as a substitute for, something which is made or occurs naturally; man-made.

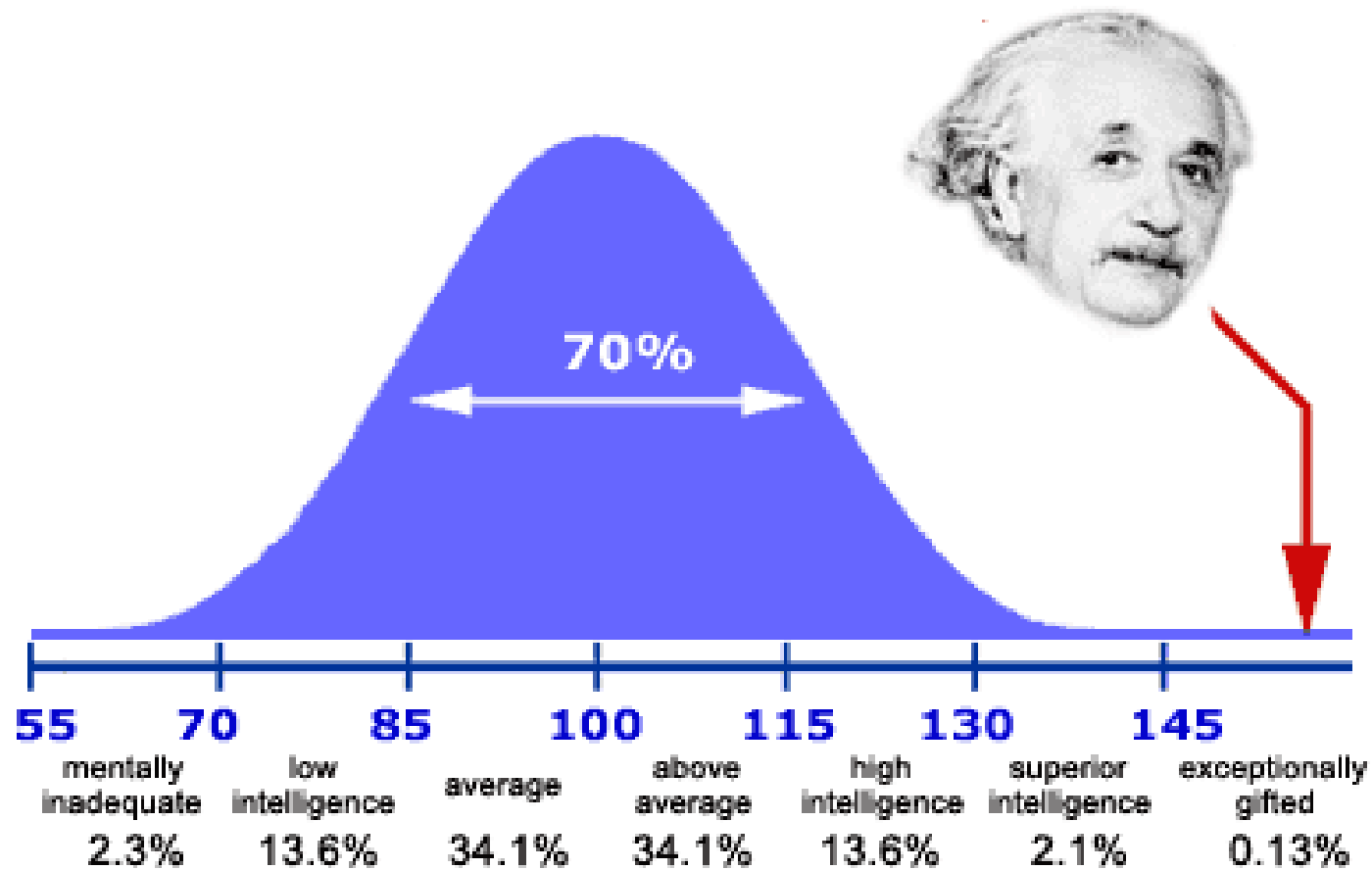
Intelligence -- The faculty of understanding; A capacity to understand; Understanding, knowledge, comprehension (of something).

Today: artificial = computation



Will AI be biological in future ?

What is intelligence ?



What is Intelligence ?

Ability to memorize ?

Ability to understand ?

Ability to recognize ?

Ability to learn ?

Ability to think ?

Ability to reason ?

Ability to judge ?

Ability to feel ?

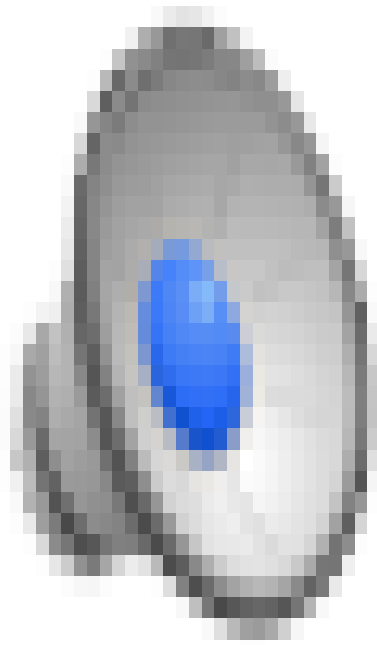
...

“Artificial intelligence (AI), the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings. The term is frequently applied to the project of developing systems endowed with the intellectual processes characteristic of humans, such as the ability to reason, discover meaning, generalize, or learn from past experience.”

-- B.J. Copeland



Automation is not intelligence



Elements of human intelligence





-- Robert J. Sternberg, "Human Intelligence", in Britannica.

"Human intelligence, mental quality that consists of the abilities to learn from experience, adapt to new situations, understand and handle abstract concepts, and use knowledge to manipulate one's environment."

"Psychologists have generally agreed that **adaptation to the environment** is the key to understanding both what intelligence is and what it does. For the most part, adaptation involves making a change in oneself in order to cope more effectively with the environment, but it can also mean changing the environment or finding an entirely new one."

"Effective adaptation draws upon a number of **cognitive processes**, such as perception, learning, memory, reasoning, and problem solving. The main emphasis in a definition of intelligence, then, is that it is not a cognitive or mental process per se but rather a selective **combination** of these processes that is **purposely directed** toward effective adaptation."

Broad technology areas of AI

1. Computer vision
2. Natural speech and language
3. Reasoning and decision making
4. Data analytics and knowledge discovery
5. Robots and autonomous systems

Computer vision related technology

- Deal with computer images/videos, 2D or 3D, capturing human or physical objects/scenes
- Perform the tasks of detection, recognition, classification, understanding, creation, rendering, transformation,

Face recognition

Object detection
/recognition

Expression
/posture/gesture
recognition

Image
segmentation

Image rendering
/modification

Image synthesis
/creation

Video
segmentation

3D
reconstruction
from 2D images

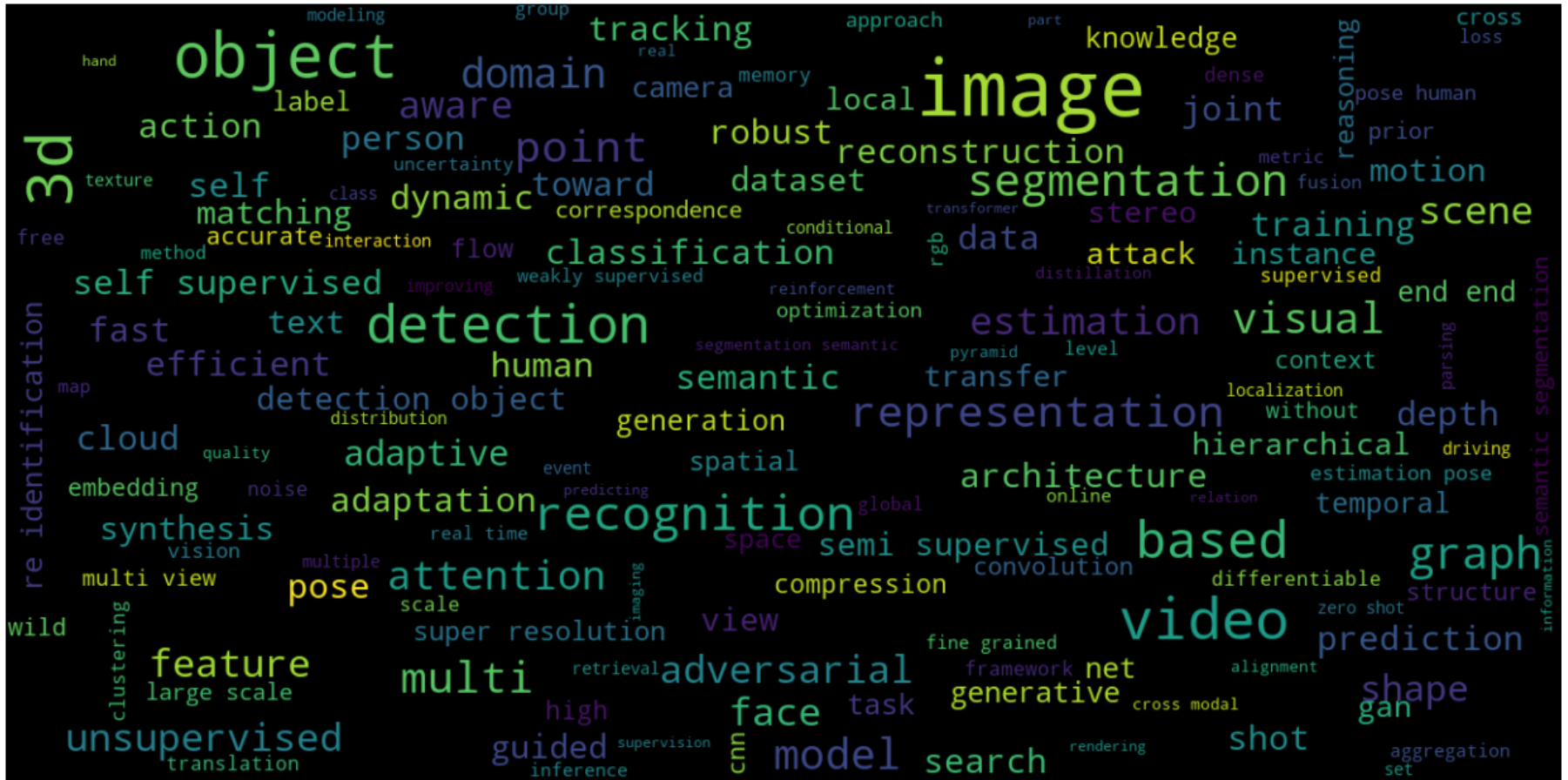
Image super-
resolution

Motion tracking
/prediction

Visual scene
understanding

Image captioning

Keywords in CV



https://github.com/hoya012/CVPR-2020-Paper-Statistics/blob/master/2020_cvpr/keyword_cloud.png

Image recognition

airplane



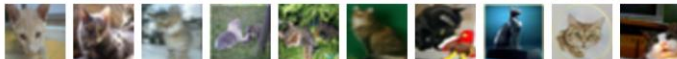
automobile



bird



cat



deer



dog



frog



horse



ship



truck



Black Jeans



Blue Dress



Blue Jeans



Blue Shirt



Red Dress



Red Shirt



Traffic scene analytics (Research of SenseTime)



Pose detection and animation



Live demo -- <https://pose-animator-demo.firebaseio.com/camera.html>

Speech and language related technology

- Deal with different kinds of languages in speech or text format.
- Perform the tasks of detection, recognition, generation, transformation, translation – all kinds of analysis and creation

Speech-to-text

Text-to-speech

Chatbot

Language
translation

Voice
conversion

Emotion
recognition

Sentiment
analysis

Concept
extraction

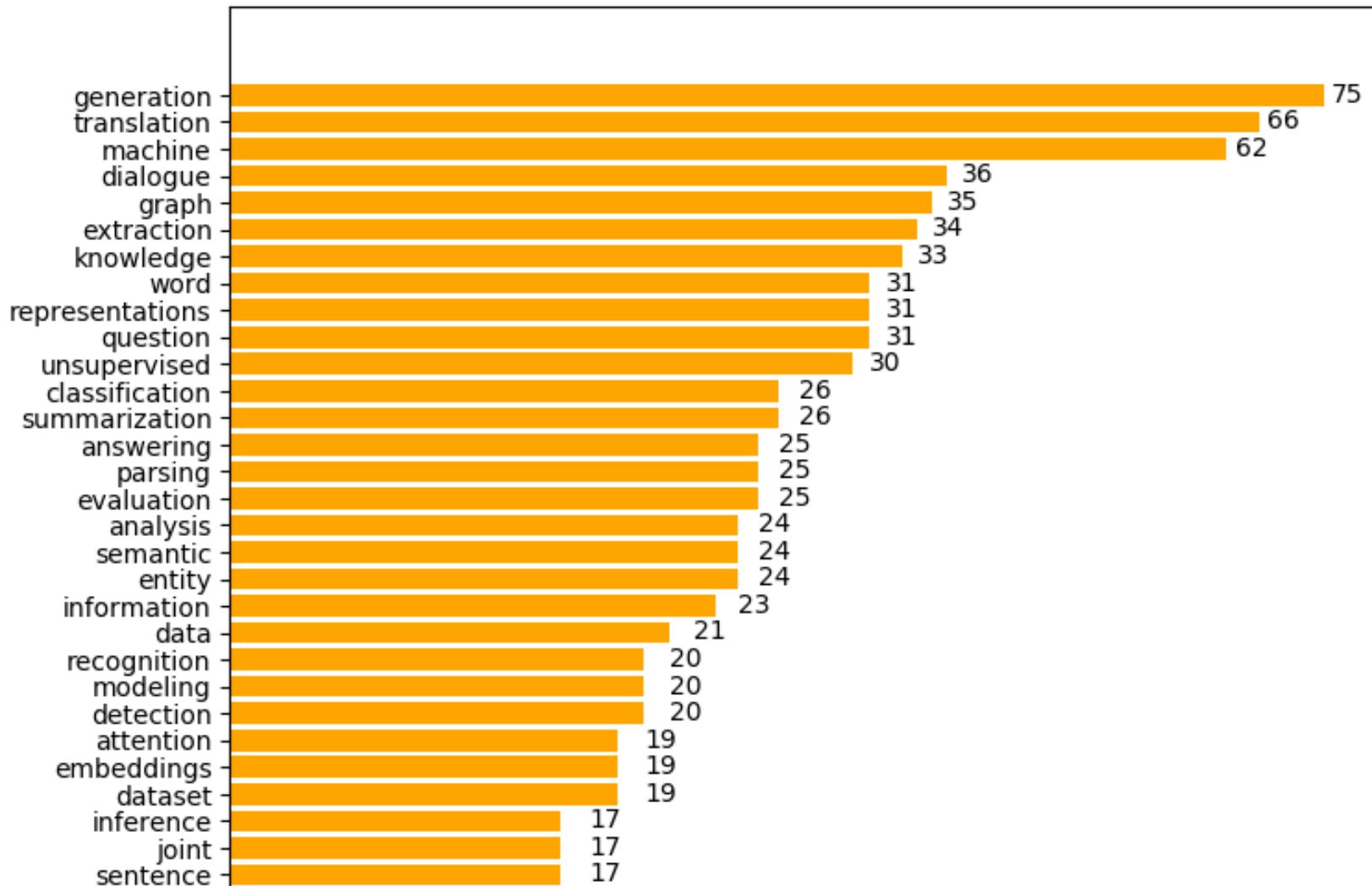
Text generation

Questioning
/answering

Text
summarization

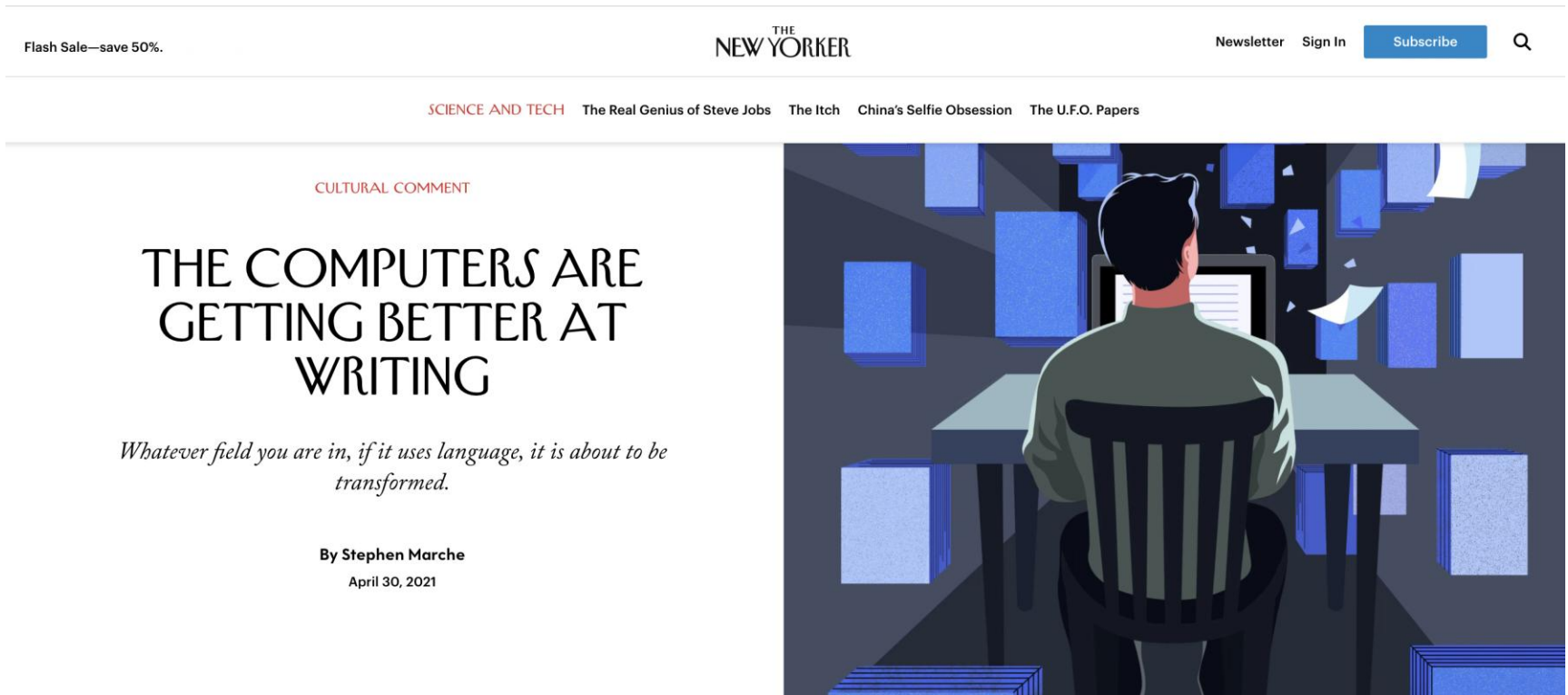
Code
generation

Top Keywords of ACL 2020 Accepted Papers



<https://github.com/hunkim/ACL-2020-Papers/blob/master/figure/keywords.png>

Sudowrite: assistive tools for writing



“Other artists have had tools like this for a long time,” he said. “There are really sophisticated tools for visual artists—whether it’s Photoshop or a 3-D tool, there’s all these things that you can apply. People who are using 3-D models aren’t using clay. They’re not using the tools of the past. They’re using really advanced tools that automate a lot of the processes. But writing has been stuck in the past. We’re not using paper and pen, but we’re not much better off than that.”



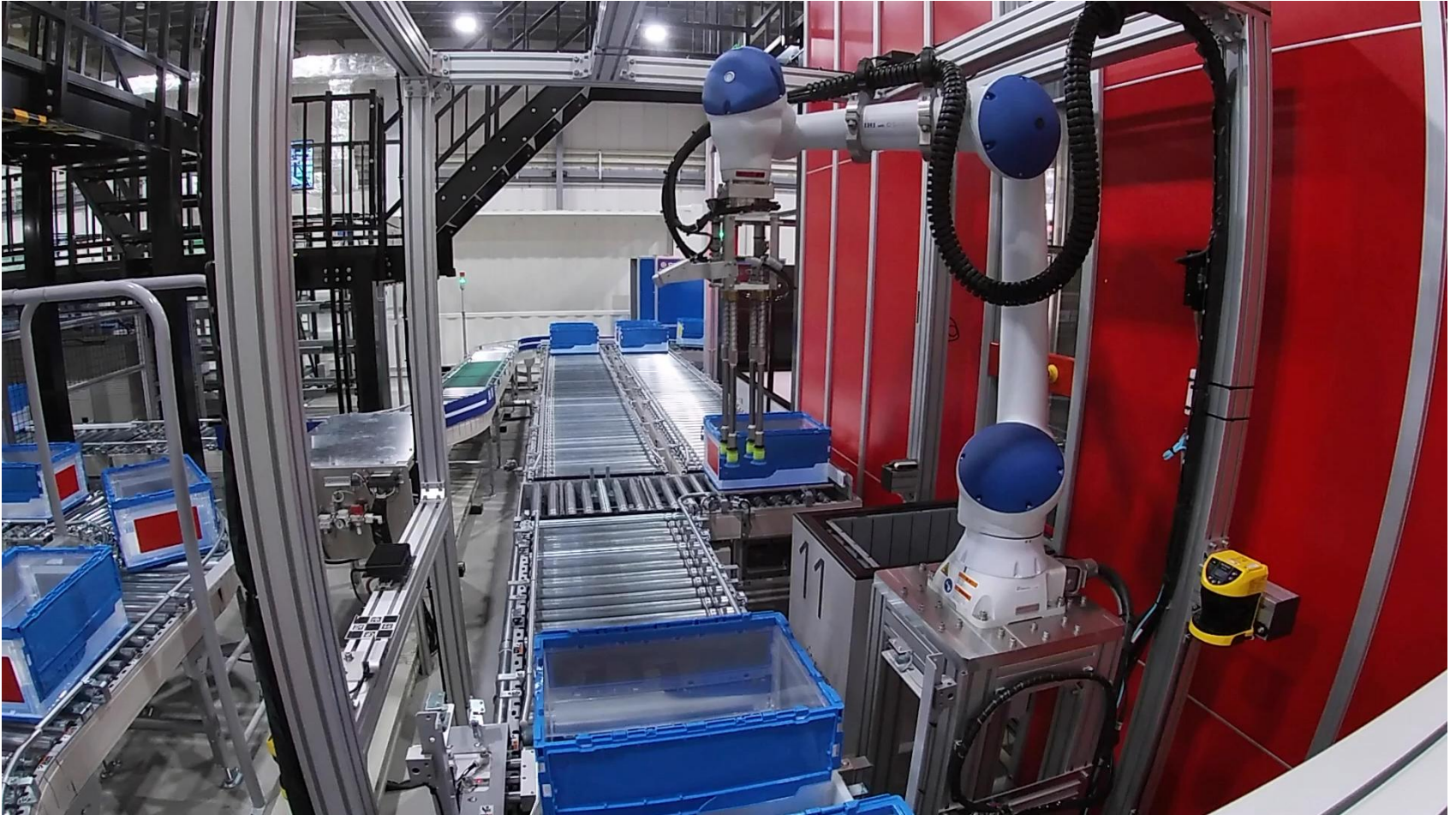
<https://www.youtube.com/watch?v=H4F3Uvnq9RM>

NLP: question and answering



<https://www.youtube.com/watch?v=DEMxVb7Q0oc>

AI-Powered Robots



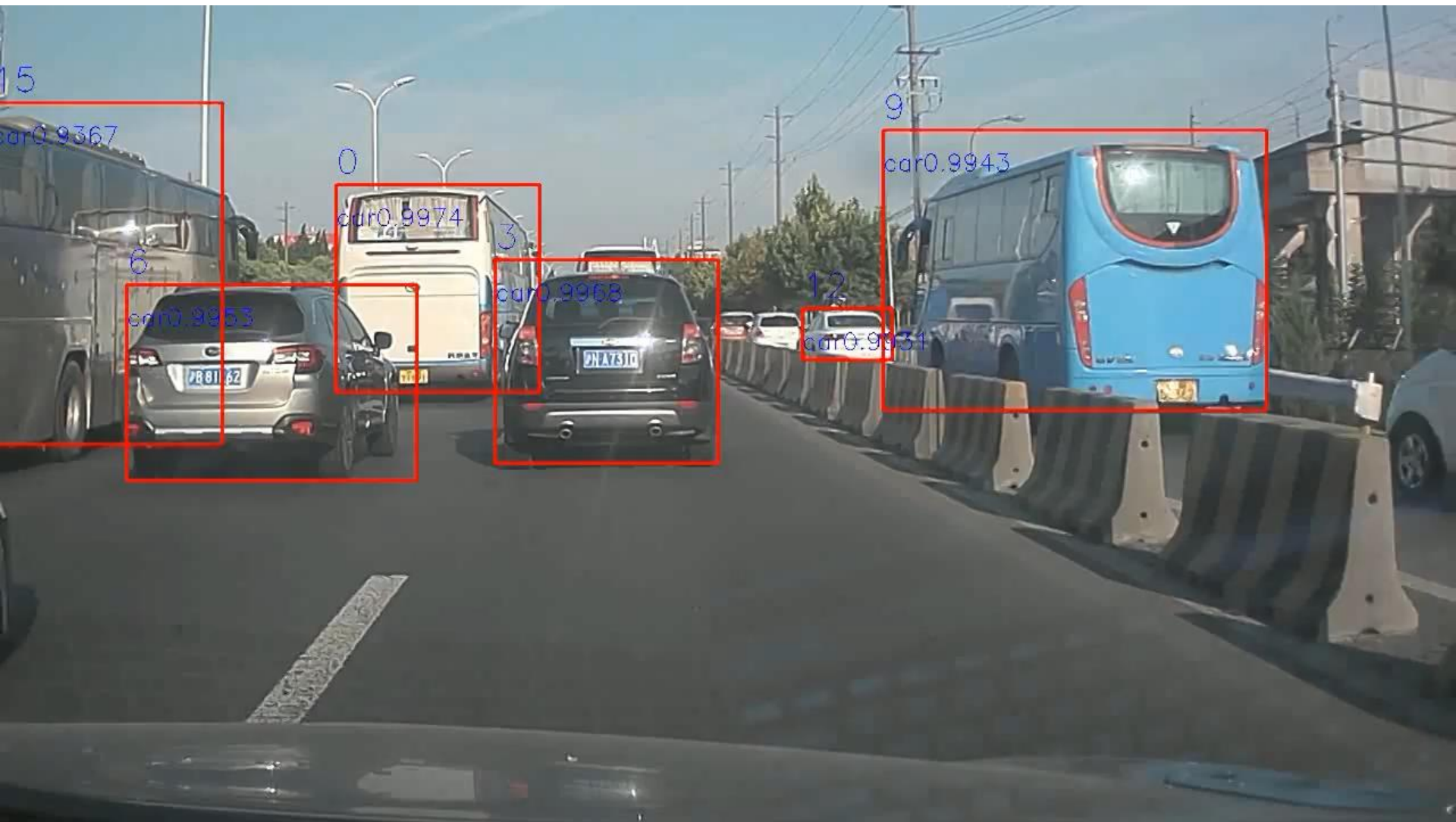
<https://www.technologyreview.com/2021/08/06/1030802/ai-robots-take-over-warehouses/>

AI Surgical Robots

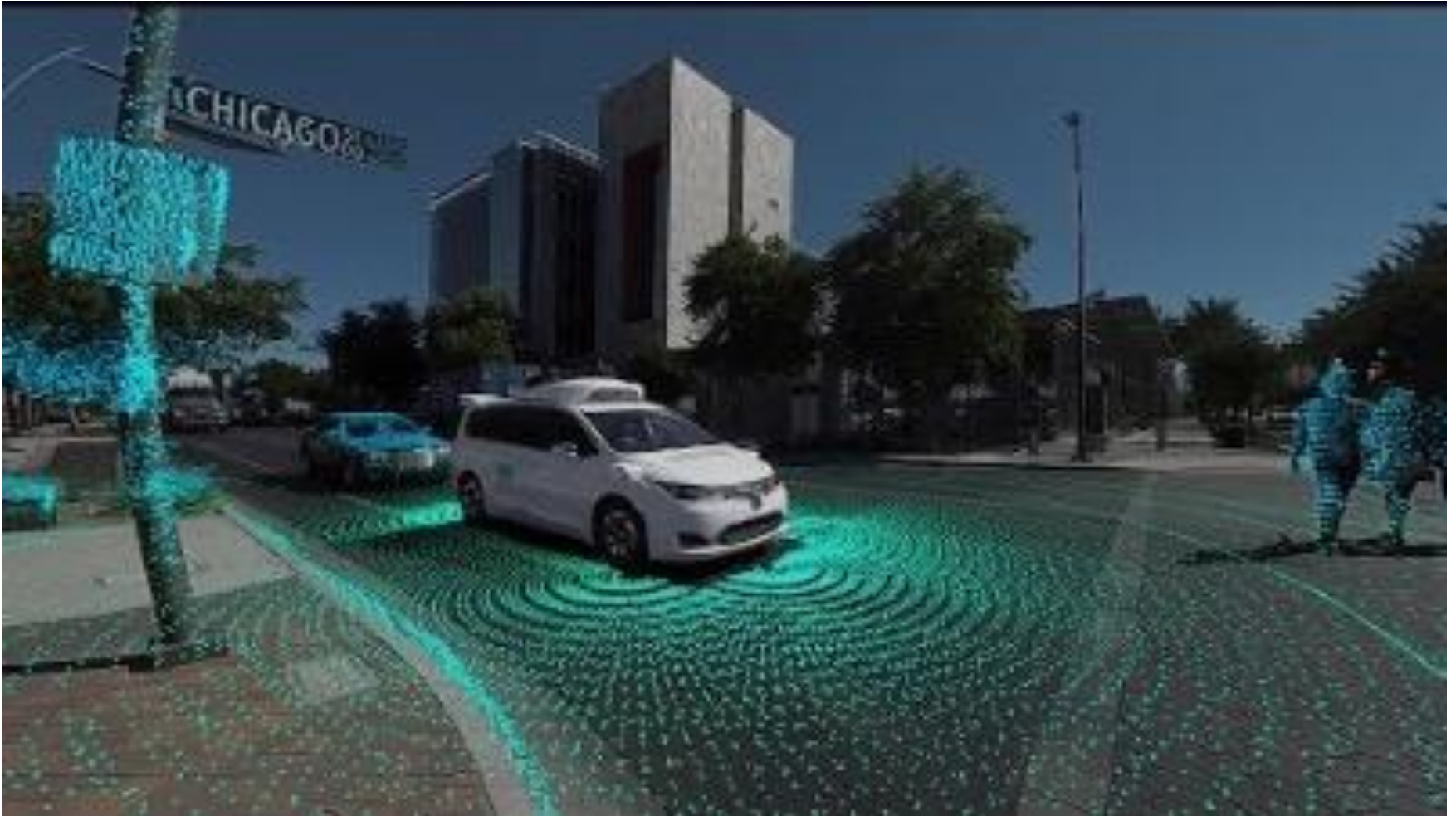


https://youtu.be/GTP7mQ-_wno

Vision in autonomous driving



Autonomous driving (Google)



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Azure Cognitive Services

Decision

Identify and analyze content within images and videos

Language

[Computer Vision](#)

Analyze content in images and video.

Speech

[Custom Vision](#)

Customize image recognition to fit your business needs.

Vision

[Face API](#)

Detect and identify people and emotions in images.

Computer Vision

Detectron2

Detectron2 is FAIR's next-generation platform for object detection and segmentation.

[GitHub](#) >[Overview](#) >

DensePose

DensePose is designed to map all human pixels of an RGB image to a 3D surface-based representation of the human body.

[GitHub](#) >[Overview](#) >

WSL Embeddings

WSL Embeddings allow experimentation with image recognition capabilities based on models trained on large datasets.

[GitHub](#) >[Overview](#) >

Language

PyText

PyText is a deep-learning based natural language processing (NLP) modeling framework built on PyTorch.

[Github](#) >[Overview](#) >

FastText

FastText is a lightweight library designed to help build scalable solutions for text representation and classification.

[Github](#) >[Overview](#) >

Translate

Translate is an open source project based on Facebook's machine translation systems.

[Github](#) >[Overview](#) >

Speech

Wav2letter

Wav2letter is an end-to-end Automatic Speech Recognition (ASR) system for transcribing speech.

[Github](#) >[Overview](#) >

Libri-light

Large dataset for the training of weakly supervised & unsupervised speech models.

[Github](#) >[Overview](#) >

Reasoning

ELF

ELF is a platform for game research that allows developers to train and test their algorithms in various game environments.

[Github](#) >[Overview](#) >

ELF OpenGo

ELF OpenGo is an AI bot from Facebook AI Research (FAIR) that has defeated world champion professional Go players.

[Github](#) >[Overview](#) >

House3D

House3D is a rich environment containing thousands of 3D scenes of visually realistic houses with full labeling.

[Github](#) >[Overview](#) >

技术能力

语音技术

图像技术

文字识别

人脸与人体识别

视频技术

AR与VR

自然语言处理

知识图谱 >

数据智能

场景方案

部署方案

查看全部AI能力

知识理解 >

实体标注 热门

汉语检索

知识问答 新品

场景方案

智能庭审辅助 邀测

行业知识图谱解决方案

AI中台 >

事件图谱 > 邀测

资讯事件判别

事件名抽取

事件属性抽取

事件相似度判别

知识中台 > 邀测

智能创作平台 >

知识图谱Schema >

| 生态合作计划

合作伙伴招募计划

合作伙伴权益与标准

| 客户案例

锤子手机—实现搜索“大爆炸”

易判科技—法律服务更智能

人民日报-助力发布“创作大脑”

紫金山—新闻媒体采编智能化

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Technical Capabilities

Based on the foundation of proprietary technologies and a core “brain” built on a deep learning platform, SenseTime has rapidly opened up AI application in multiple vertical scenarios.



SenseCore AI Infrastructure



Intelligent Automobile Technology



General Artificial Intelligence



Augmented Reality



AI-enabled Content Generation



Decision Intelligence



AI-enabled Content Enhancement



Medical Image Analysis



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Scientific advances, real world benefits

On this page



Understanding protein folding

Proteins are complex molecules that are essential to life. Each has its own unique 3D shape that determines how it works and what it does.

Knowing how proteins fold could help scientists understand the biological processes of every living thing. To accelerate progress, we created [AlphaFold](#), a system which accurately predicts the shape of proteins. This research has enormous potential in every field of biology, from helping tackle disease and quickly finding new medicines to unlocking the mysteries of how life itself works.

Identifying eye disease faster

We partnered with Moorfields Eye Hospital to develop faster ways of identifying, and better ways of understanding, common eye diseases from routine scans.

Over 100 million people are affected by diabetic retinopathy or age-related macular degeneration. These conditions can cause permanent sight loss unless they're treated quickly. The results, which were published in [Nature Medicine](#), showed that our AI system could recommend patient referrals as accurately as world-leading expert doctors for over 50 sight-threatening eye diseases. More recently, we showed that [our system can predict](#) whether a patient will develop a more severe form of age-related macular degeneration months before it happens—paving the way for future research in sight-loss prevention.

[API](#)[Research](#)[Blog](#)[About](#)

OpenAI is an AI research and deployment company. Our mission is to ensure that artificial general intelligence benefits all of humanity.

How it works

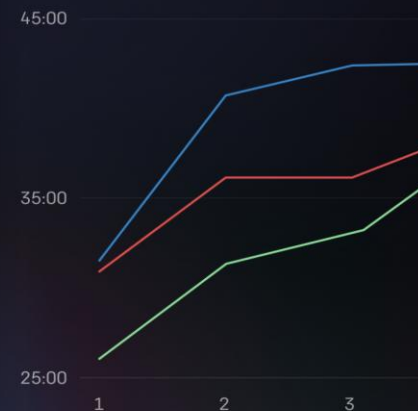
OpenAI Codex is an AI system that can autocomplete code. When the Codex Challenge opens, you'll be presented with five Python programming puzzles. You can compete against the Internet and Codex itself to solve them, or just go at your own pace and enjoy the problems.

Finish time
10:28:34 AM

#96

👤 TOP 100

Problem 1	04:56
Problem 2	05:34
Problem 3	06:02
Problem 4	05:59
Problem 5	06:03
Time spent	28:34



Auto-programming



Make it be the size of the rocketship times 0.75



```
text.style.left =
rocketship.offsetLeft + 'px';
text.style.top =
rocketship.offsetTop + 'px';

document.body.appendChild(text)
}
speed = 10;
setInterval(function() {
    speed = 1;

    document.body.removeChild(text)
}, 100);
}
```

→ Now add an image of an
asteroid.
Steps /> cd /home/username/full
/172108/asteroid.jpg?
w=1000&h=1000&w=1000&h=1000
setInterval(function() {
var asteroid =
document.createElement('img');
asteroid.src =
http://172108/asteroid.jpg?
w=1000&h=1000&w=1000&h=1000
setInterval(function() {
document.body.appendChild(asteroid);
}

<https://www.youtube.com/watch?v=Zm9B-DvwOgw>

Robot Auto-Learning



CORTIS **REDRED**

While China dominates humanoid robotics headlines, a Korean firm showcased a humanoid learning complex motions through an open-source AI framework.

TRENDING

LATEST

F-22 Raptor: Pratt
& Whitney's F119

ANI and AGI

The buzz word “AI” is two different ideas.

Artificial Narrow Intelligence (ANI)

- Using AI to do one specific and clearly defined task
- Progressing rapidly. Examples we quoted above are all ANI
- Very valuable and impactful to individuals and society

Artificial General Intelligence (AGI)

- Having AI to do anything human can do
- So far very little progress, if any at all
- Unfortunately creating worries and irrational fears
- Not the focus of this course: not much to talk about

AI, ML, DL and Data Science

